

034IN - FONDAMENTI DI AUTOMATICA
- FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION
A.Y. 2023/2024

June 21 2024

First and Last Name:

Group: Group A

Exercise: Exercise 1 - 2 Questions

Available Time Frame: 60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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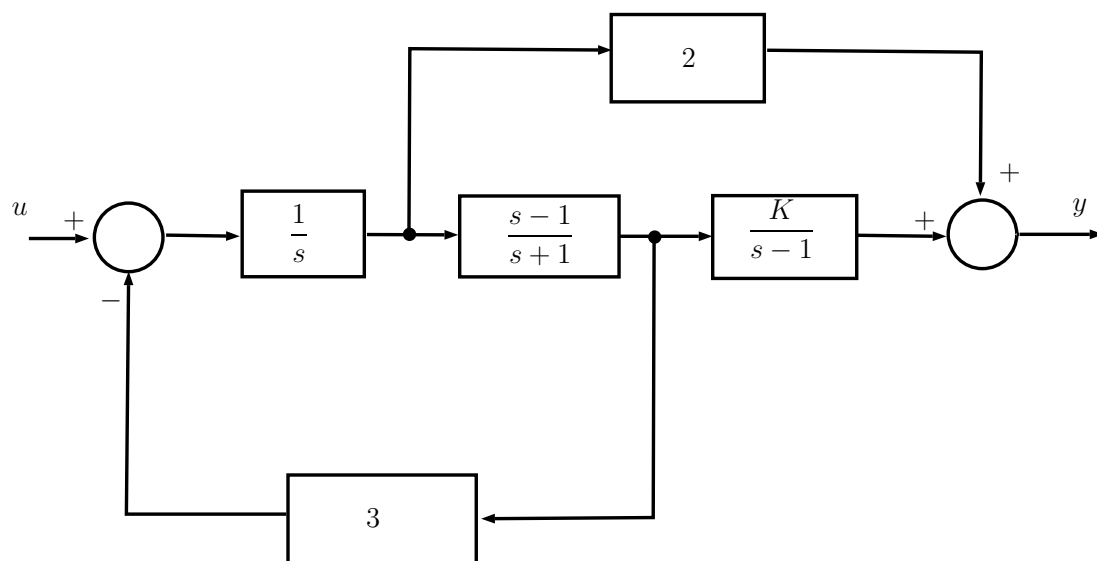
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Exercise 1

Consider the block diagram in the figure:



Question 1.1. Determine a **realisation using state equations** for the overall system.

Question 1.2. Analyse the overall system's stability as the real parameter K changes.